Primer: CCP-NPL Estimation

Hotz and Miller (1993), Aguirregabiria and Mira (2002)

econirl package documentation

1 Problem and Motivation

The Nested Fixed Point (NFXP) estimator of Rust (1987) is statistically efficient but requires solving the Bellman equation at every outer optimization step. The inner loop anchors gradient computation: you cannot evaluate the log-likelihood without first computing the value function. This creates a fundamental asymmetry between estimation and identification — even a researcher who only wants a quick, consistent preliminary estimate must invoke the full dynamic programming machinery.

Hotz and Miller (1993) showed this is unnecessary. Under additive separability and IID private shocks, the observed choice probabilities $P(a|s)$ contain all the information needed to recover continuation values without solving any fixed point. Their inversion lemma maps CCPs directly to value functions via a single matrix solve, replacing the inner loop with one $O(|\mathcal{S}|^3)$ matrix inversion. A first-step consistent estimator follows immediately from empirical choice frequencies.

Aguirregabiria and Mira (2002) showed how to recover full MLE efficiency from this starting point. Iterating the Hotz-Miller mapping in CCP space — the Nested Pseudo-Likelihood (NPL) algorithm — converges geometrically to the MLE fixed point. Five NPL iterations suffice in practice. This paper established the zero Jacobian property: at the NPL fixed point, the estimator is first-order insensitive to CCP estimation error, delivering the same asymptotic efficiency as NFXP without its structural starting requirement.

CCP estimation is the only method in the EconIRL package that extends naturally to dynamic games. NFXP requires computing Nash equilibria in the inner loop and is therefore infeasible for games with more than two players.

2 Model and Notation

Table 1: Common DDC notation shared across all primers.

Symbol	Name	Definition
$s \in \mathcal{S}$	State	Discrete or continuous
$a \in \mathcal{A}$	Action	Finite: $\lvert\mathcal{A}\rvert = J$
s'	Next state	Drawn from $P(s' \mid s, a)$
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale ($= 1$ by convention)
$\theta \in \Theta \subset \mathbb{R}^{d_\theta}$	Structural parameters	Unknown
$\phi(s, a)$	Feature vector	Known functions of (s, a), dimension K
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$ (linear-in-parameters)
$P(s' \mid s, a)$	Transition kernel	Known or estimated from data
$Q(s, a)$	Choice-specific value	$u(s, a; \theta) + \beta \sum_{s'} P(s'\|s, a)\, V(s')$
$V(s)$	Integrated value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a \mid s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$ from one agent
$\mathcal{D}$	Panel data	Collection of N observed trajectories

We work within the DDC framework of Rust (1987). An agent in state $s \in \mathcal{S}$ chooses action $a \in \mathcal{A}$ to maximize discounted expected utility. Flow utility is $u(s, a; \theta) + \sigma\epsilon_a$ where ϵ_a is Type I extreme value with scale $\sigma > 0$. Transitions follow $P(s'|s, a)$ with discount factor $\beta \in [0, 1)$.

The key objects are as follows.

Symbol	Name	Definition
$e(a, s)$	Emax correction	$\gamma_{\text{Euler}} - \log \pi(a\|s)$, where $\gamma_{\text{Euler}} \approx 0.5772$
F_π	Policy-weighted transitions	$F_\pi[s, s'] = \sum_a \pi(a\|s)\, P(s'\|s, a)$
W_ϕ	Augmented feature matrix	$(I - \beta F_\pi)^{-1} \sum_a \pi(a) \odot \phi(a)$, shape $(\|\mathcal{S}\|, K)$
W_e	Augmented emax vector	$(I - \beta F_\pi)^{-1} \sum_a \pi(a) \odot e(a)$, shape $(\|\mathcal{S}\|,)$
$\tilde{\phi}(s, a)$	Augmented features	$\phi(s, a) + \beta \sum_{s'} P(s'\|s, a)\, W_\phi(s')$
$\tilde{e}(s, a)$	Augmented emax	$\beta \sum_{s'} P(s'\|s, a)\, W_e(s')$

3 Core Theory

3.1 The Hotz-Miller Inversion Lemma

Under additive separability (AS), conditional independence (CI), and IID extreme value errors (EV), the integrated value function at state s under policy π can be written as

$$\bar{v}(s) = \sum_a \pi(a|s)\big[u(s, a; \theta) + e(a, s)\big] + \beta \sum_{s'} F_\pi(s, s')\, \bar{v}(s'). \tag{1}$$

The emax correction $e(a, s) = \gamma_{\text{Euler}} - \log \pi(a|s)$ is entirely determined by the CCPs and the known distributional assumption on ϵ. No knowledge of θ or V is needed to compute it.

Rearranging Equation (1) in matrix form gives

$$\bar{v} = (I - \beta\, F_\pi)^{-1} \left[\sum_a \pi(a) \odot \big(u(a; \theta) + e(a)\big) \right]. \tag{2}$$

The matrix $(I - \beta F_\pi)$ is invertible because $\beta < 1$ and F_π is stochastic. This is the Hotz-Miller inversion: the continuation value is recovered by a single matrix inversion without any iteration.

3.2 Choice-Specific Values from CCPs

For linear utility $u(s, a; \theta) = \theta^\top \phi(s, a)$, the choice-specific value function separates into

$$v(s, a; \theta) = \theta^\top \tilde{\phi}(s, a) + \tilde{e}(s, a). \tag{3}$$

The augmented features $\tilde{\phi}(s, a)$ and emax correction $\tilde{e}(s, a)$ depend only on the current CCPs π, not on θ. This separation is the implementation key: given fixed CCPs, the matrices W_ϕ and W_e are computed *once* per NPL step, after which the pseudo-likelihood optimization in θ is a standard logit on augmented features with no further matrix inversion.

3.3 NPL Convergence (Aguirregabiria-Mira 2002)

Aguirregabiria and Mira (2002) define the NPL mapping $\Psi : \pi \mapsto \pi'$ as follows. Given CCPs $\pi^{(k)}$, compute augmented features $\tilde{\phi}, \tilde{e}$, maximize the pseudo-log-likelihood

$$\hat{\theta}^{(k)} = \arg\max_\theta \sum_{i=1}^{N} \log \frac{\exp\big(v(s_i, a_i;\, \theta, \pi^{(k)})/\sigma\big)}{\sum_{a'} \exp\big(v(s_i, a';\, \theta, \pi^{(k)})/\sigma\big)}, \tag{4}$$

then update $\pi^{(k+1)}(a|s) = \text{softmax}\big(v(s, a;\, \hat{\theta}^{(k)}, \pi^{(k)})/\sigma\big)$. A fixed point of Ψ is an NPL equilibrium.

The **zero Jacobian property** ensures that at the fixed point, the estimator is first-order insensitive to CCP estimation error. This means the NPL fixed point achieves the same asymptotic distribution as NFXP-MLE, despite starting from noisy empirical CCPs.

3.4 Identification

CCP estimation inherits the same identification structure as NFXP. The parametric restriction $u = \theta^\top \phi(s, a)$ with $K < |\mathcal{S}|(|\mathcal{A}| - 1)$ resolves reward shaping ambiguity. The discount factor β and scale σ are typically fixed; Magnac and Thesmar (2002) showed that β and the reward level are not jointly identified without further restrictions.

4 Estimation Algorithm

Algorithm 1: CCP-NPL Estimation (Hotz and Miller, 1993; Aguirregabiria and Mira, 2002)

Input: Panel $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N$, features $\phi(s, a)$, transitions $P(s'|s, a)$, discount β, scale σ, NPL iterations K

`// Step 1: Estimate CCPs from data (non-parametric first step)`
for *each state* s **do**
  $\hat{\pi}^{(0)}(a|s) = N(s, a)/N(s)$ `// frequency estimator`
  if $N(s) < n_{\min}$ **then** $\hat{\pi}^{(0)}(a|s) = 1/|\mathcal{A}|$ `// fallback to uniform`
end
Apply Laplace smoothing: $\hat{\pi}^{(0)} \leftarrow (\hat{\pi}^{(0)} + \delta)/(1 + |\mathcal{A}|\delta)$

`// Step 2: NPL loop`
for $k = 0, 1, \ldots, K - 1$ **do**
  `// 2a. Compute emax corrections and policy-weighted transitions`
  $e^{(k)}(a, s) = \gamma_{\text{Euler}} - \log \hat{\pi}^{(k)}(a|s)$
  $F_\pi^{(k)}[s, s'] = \sum_a \hat{\pi}^{(k)}(a|s)\, P(s'|s, a)$
  `// 2b. ONE matrix inversion per NPL step --- the implementation key`
  $W_\phi = (I - \beta F_\pi^{(k)})^{-1} \sum_a \hat{\pi}^{(k)}(a) \odot \phi(a)$
  $W_e = (I - \beta F_\pi^{(k)})^{-1} \sum_a \hat{\pi}^{(k)}(a) \odot e^{(k)}(a)$
  `// 2c. Augmented features --- fixed for this NPL step`
  for *each* (s, a) **do**
    $\tilde{\phi}(s, a) = \phi(s, a) + \beta \sum_{s'} P(s'|s, a)\, W_\phi(s')$
    $\tilde{e}(s, a) = \beta \sum_{s'} P(s'|s, a)\, W_e(s')$
  end
  `// 2d. Pseudo-LL maximization: standard logit on` $\tilde{\phi}$`, no further inversion`
  $v(s, a; \theta) = \theta^\top \tilde{\phi}(s, a) + \tilde{e}(s, a)$
  $\hat{\theta}^{(k)} = \arg\max_\theta \sum_{i=1}^N \log \operatorname{softmax}\big(v(s_i, \cdot; \theta)/\sigma\big)_{a_i}$ `// L-BFGS-B`
  `// 2e. Update CCPs`
  $\hat{\pi}^{(k+1)}(a|s) = \operatorname{softmax}\big(v(s, a; \hat{\theta}^{(k)})/\sigma\big)$
  if $\|\hat{\theta}^{(k)} - \hat{\theta}^{(k-1)}\| < \varepsilon$ **then break**
end

`// Step 3: Standard errors`
Compute numerical Hessian H at $\hat{\theta}$ via the full Bellman-constrained log-likelihood
$\text{SE}(\hat{\theta}) = \sqrt{\operatorname{diag}((-H)^{-1})}$

Output: $\hat{\theta}$, V^*, $\hat{\pi}$, standard errors

Implementation notes. The econirl implementation caches W_ϕ and W_e at the start of each NPL step and passes the precomputed augmented features $\tilde{\phi}$, $\tilde{e}$ to the logit optimizer. This means each gradient evaluation in Step 2d costs $O(N \cdot |\mathcal{A}| \cdot K)$ rather than $O(|\mathcal{S}|^3)$, reducing per-gradient cost by orders of magnitude for large $|\mathcal{S}|$. The full matrix inversion cost $O(|\mathcal{S}|^3)$ is paid exactly once per NPL step. The numerical Hessian in Step 3 uses the full Bellman-constrained log-likelihood rather than the pseudo-likelihood with fixed CCPs, because the pseudo-likelihood

Hessian is locally flat in directions that primarily affect CCPs, producing spurious rank deficiency.

5 Results

Table 2: NPL convergence to MLE on Rust bus (90 states, $\beta = 0.9999$, 20,000 observations). True $\theta_c = 0.001$, $RC = 3.0$. NFXP is the MLE reference. BC policy accuracy is 98.9%.

Metric	K=1 (HM)	K=2	K=3	K=5	NFXP
$\hat{\theta}_c$	-0.000830	0.000864	0.001073	0.001225	0.001231
$\hat{RC}$	2.8568	2.9872	3.0008	3.0101	3.0115
Log-likelihood	-4285.25	-4266.27	-4263.36	-4263.20	-4263.20
—LL gap—	22.0503	3.0726	0.1611	0.0004	—
Cosine sim.	1.0000	1.0000	1.0000	1.0000	1.0000
SE(θ_c)	—	—	—	0.000305	0.000309
SE(RC)	—	—	—	0.051293	0.051597
Time (s)	1.5	1.2	1.8	7.5	3.9

Table 3: Sensitivity to starting values on Rust bus (90 states, $\beta = 0.9999$). NFXP outer iterations and log-likelihood vary with the starting point. CCP K=5 produces identical estimates regardless of starting theta: $\hat{\theta}_c$ range 0.0000000, $\hat{RC}$ range 0.0000.

Starting point	NFXP			CCP K=5		
	Iter.	$\hat{\theta}_c$	$\hat{RC}$	Iter.	$\hat{\theta}_c$	$\hat{RC}$
Default (data-driven)	12	0.001231	3.0115	5	0.001225	3.0101
Near truth $(0.001, 3.0)$	12	0.001231	3.0115	5	0.001225	3.0101
Bad 1: $(1.0, 0.1)$	31	0.001231	3.0115	5	0.001225	3.0101
Bad 2: $(0.0001, 50)$	55	0.001231	3.0115	5	0.001225	3.0101
Bad 3: $(10^{-5}, 10^{-5})$	12	0.001231	3.0115	5	0.001225	3.0101

* Did not converge — result at iteration limit.

NPL convergence to MLE (Table 2). Starting from pure empirical choice frequencies (K=1, Hotz-Miller), the LL gap to NFXP is 22.05 nats — consistent but not efficient. By K=3 the gap has collapsed to under 0.2 nats and the parameter estimates are essentially recovered. At K=5 the gap is 0.0004, which is machine precision relative to the problem scale. This is the zero Jacobian property in action: each NPL iteration removes the first-order sensitivity to CCP estimation noise, and five steps suffice to eliminate it entirely.

K=1 also runs in 1.5s versus NFXP's 3.9s on the same problem, a roughly 2.5× advantage. The gain is structural: NFXP must solve the Bellman equation at every outer gradient evaluation, while K=1 pays one $O(|\mathcal{S}|^3)$ matrix inversion followed by a plain logit. For large state spaces the per-gradient cost gap widens further.

Initialization robustness (Table 3). The second table exposes a qualitative difference between the two estimators. NFXP requires a parametric starting point θ_0; from a well-specified start (near the truth) it converges in 12 outer iterations, but from a misspecified start (e.g. $RC = 50$) it requires 55 outer iterations — each involving a full inner Bellman solve. The researcher must either guess a reasonable θ_0 or run a preliminary screening pass.

CCP K=5 is invariant to θ_0 by construction. Step 1 reads CCPs directly from empirical frequencies; the logit in Step 2d is convex and converges to the same global optimum regardless of where it starts. Across all five starting points the NPL fixed point is identical to seven decimal places (Table 3). This means CCP does not require a preliminary screening pass or a good initial guess — it is self-initializing.

Code. `ccp_run.py` generates Tables 2 and 3. Estimator: `src/econirl/estimation/ccp.py`. Behavioral cloning achieves 98.9% policy accuracy (log-likelihood -4249.41), confirming that CCP learns from MDP structure beyond empirical action frequencies.

```
.venv/bin/python papers/econirl_package/primers/ccp/ccp_run.py
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References

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